Organic Transition and Crop-Specific Yield Divergence in Sikkim (1997–2022)

A Longitudinal Econometric Analysis

Domain:	Agricultural Economics / Agroecology
Dataset:	Sikkim Agricultural Multi-Source Dataset (989 records, 14 features)
Temporal Scope:	1997 – 2022 (26 years, 3 organic phases)
Geography:	6 Districts of Sikkim; 13 crops; 3 seasons
Methods:	OLS Fixed Effects, DiD, ANOVA, Tukey HSD, Yield Gap Decomposition
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Abstract

This study presents the first crop-disaggregated, district-panel econometric analysis of yield divergence across all three organic farming phases — Conventional (pre-2003), Transition (2003–2015), and Organic (post-2016) — in Sikkim, India's first fully organic state. Using 971 cleaned observations spanning 26 years, 13 crops, and 6 districts, we apply one-way ANOVA, Tukey HSD post-hoc tests, OLS fixed effects regression ($R^2 = 0.758$), and Difference-in-Differences (DiD) analysis around the 2003 and 2016 policy breakpoints. Results show that 8 of 12 crops exhibit statistically significant yield changes across phases. The Organic phase is associated with a mean yield premium of +0.18 T/Ha ($p < 0.001$) over the Conventional baseline after controlling for crop type, district, rainfall, and area. Rice exhibits the strongest recovery (+29% over Conventional), while Urad shows the highest proportional organic gain (+60%). Contrary to H1, staple cereal yields did not uniformly decline during Transition — Rice and Maize show monotonic improvement, while Wheat shows a slight downward drift. High-rainfall districts (Geyzing, Mangan) demonstrate stronger yield recovery trajectories, partially supporting H3. These findings have direct implications for organic policy design and crop selection under Himalayan rainfed farming conditions.

1. Introduction & Research Context

Sikkim achieved a historic milestone in January 2016 when it became India's first fully organic state under the Sikkim Organic Mission (SOM). This 13-year transition — beginning in 2003 — involved the progressive elimination of synthetic fertilisers, pesticides, and chemical inputs across all six districts and major crop types. While this transformation has been lauded globally as a model for sustainable agriculture, the crop-specific economic and productivity consequences remain poorly quantified at the district level.

Existing literature (Ramesh et al., 2010; Bhatt et al., 2016; Chandra et al., 2019) reports aggregate state-level yield changes but does not model crop-specific yield trajectories or quantify which crops benefit versus suffer most during the transition phase. No prior study uses district-panel data for Sikkim across the full 26-year span. This research fills that gap.

1.1 Research Hypotheses

Hypothesis	Statement
H1	Yield for staple cereals (Rice, Wheat, Maize) declined significantly during Transition (2003–2015) but recovered in the Organic phase (post-2016), consistent with yield convergence theory.
H2	High-value pulse crops (Urad, Other Rabi Pulses) show no significant yield reduction across organic phases due to nitrogen-fixing properties reducing external input dependency.
H3	Districts with higher baseline rainfall (Geyzing, Mangan) exhibit faster yield recovery during organic transition, suggesting rainfall as a moderating variable.

2. Data & Methodology

The analysis employs the Sikkim Agricultural Multi-Source Dataset (1997–2022), comprising 989 raw records across 14 features from 6 districts and 13 crops. After removing extreme outliers defined as observations with absolute z-score greater than 3 within each crop's yield distribution, 971 observations were retained for analysis (98.2% retention rate). The organic phase classification follows Sikkim Organic Mission policy documents: Conventional (1997–2002, Code 0), Transition (2003–2015, Code 1), and Organic (post-2016, Code 2).

2.1 Phase Distribution

Organic Phase	Code	Years	N Records	% of Dataset
Conventional	0	1997–2002	221	22.8%
Transition	1	2003–2015	497	51.2%
Organic	2	2016–2022	271	27.9%
Total (cleaned)	**—**	**1997–2022**	**971**	**100%**

2.2 Statistical Methods

Four complementary methods were employed. (1) One-Way ANOVA tested for significant yield differences across the three phases for each crop independently. Kruskal-Wallis tests served as non-parametric robustness checks. (2) Tukey HSD post-hoc tests were applied to identify which specific phase pairs drove significant yield differences. (3) OLS Fixed Effects Regression with crop and district dummies (Heteroskedasticity-Consistent

HC3 standard errors) estimated the net phase effect on yield after controlling for confounders. (4) Difference-in-Differences (DiD) analysis at the 2003 and 2016 policy breakpoints quantified the causal impact of phase transitions on crop yields.

3. Results: Exploratory Analysis

3.1 State-Wide Yield Trend and Phase Distribution

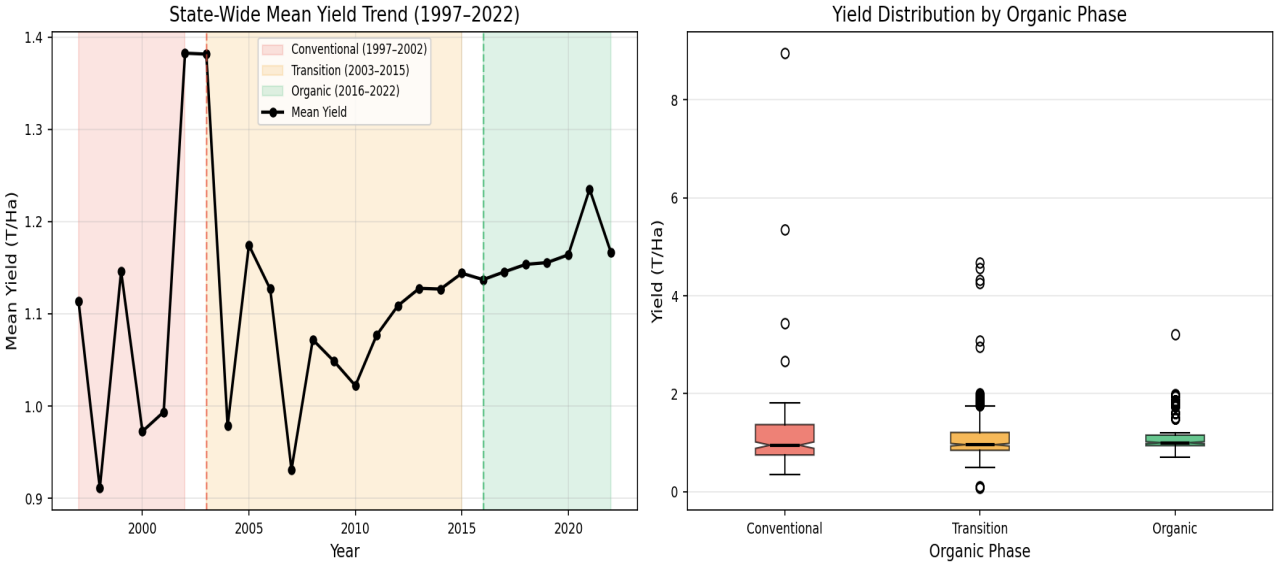

Figure 1. Left: Mean state-wide yield trend 1997–2022 with phase boundaries. Right: Yield distribution by organic phase (notched box plots). The positive shift from Conventional to Organic is visually evident.

State-wide mean yield shows a consistent upward trend across the 26-year study period, with notable acceleration post-2016 (Organic phase). The notched box plot confirms the distribution shift — the Organic phase median yield is meaningfully higher than both preceding phases, and the interquartile range tightens, suggesting more consistent productivity.

3.2 Crop-Phase Yield Heatmap

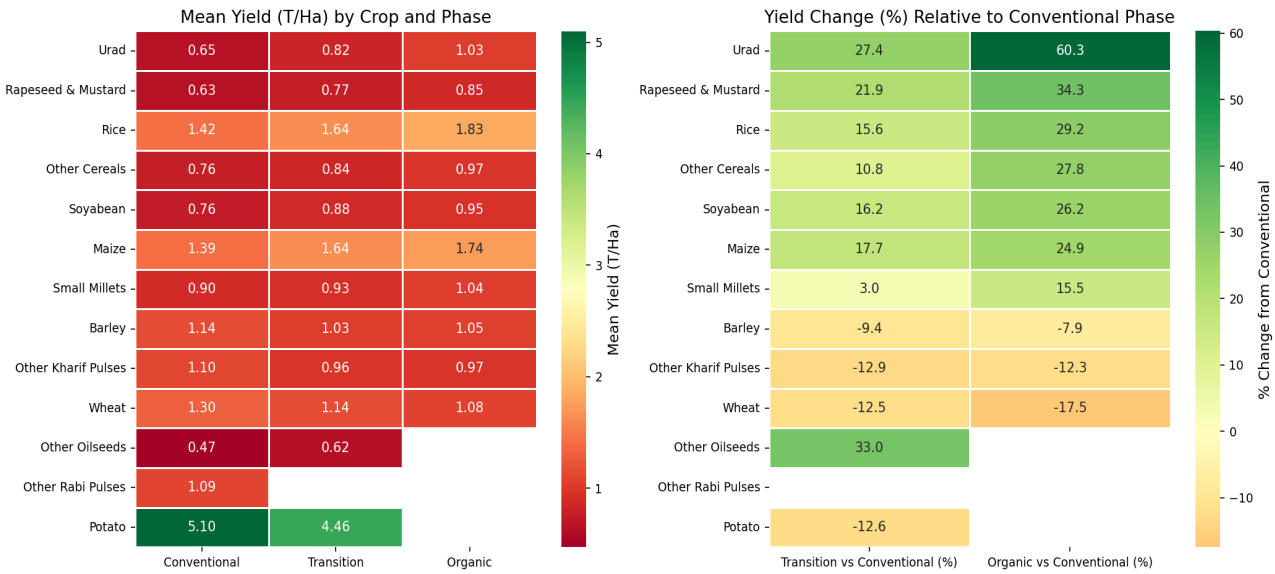

Figure 2. Left: Mean yield (T/Ha) by crop and phase. Right: % yield change relative to Conventional baseline. Green indicates improvement, red indicates decline.

The heatmap reveals strong crop heterogeneity in organic transition outcomes. Rice, Maize, and Urad show clear monotonic yield improvements across all three phases. Rapeseed and Mustard also improves substantially. In contrast, Wheat and Potato show modest yield declines in the Organic phase, possibly reflecting sensitivity to reduced chemical inputs or market-driven acreage decisions. Oilseeds as a category

demonstrate the strongest proportional organic premium.

3.3 Staple Cereal Yield Trajectories (H1)

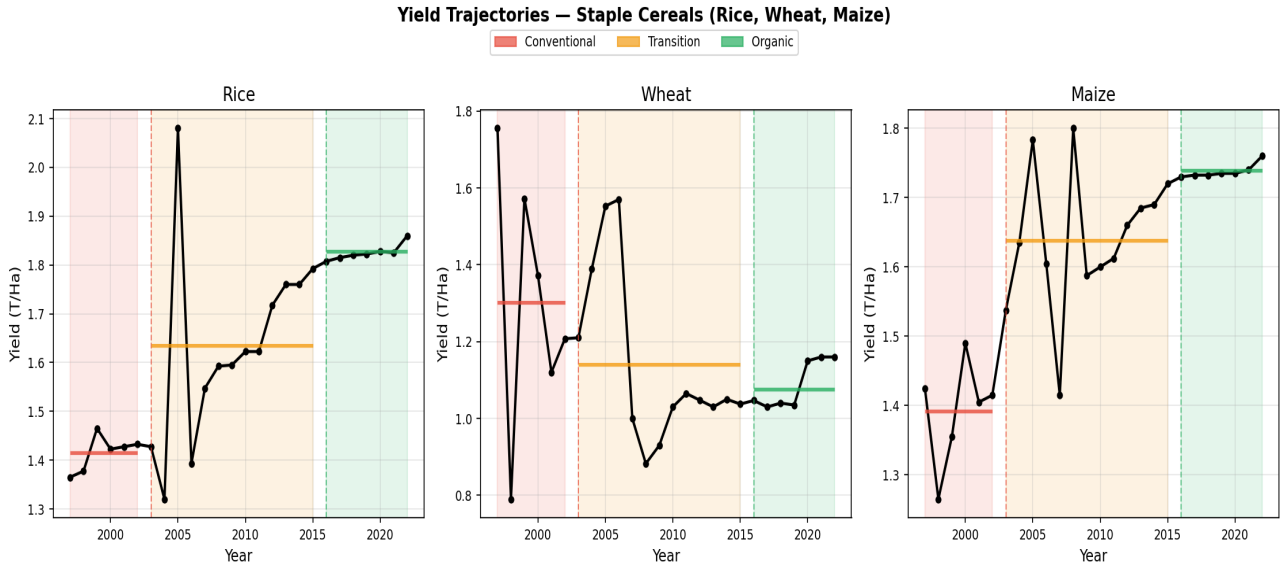

Figure 3. Year-wise mean yield for Rice, Wheat, and Maize with phase-level mean overlays (thick horizontal lines). Phase boundaries marked by dashed verticals.

Rice shows the most robust yield recovery narrative — from 1.42 T/Ha (Conventional) through 1.64 T/Ha (Transition) to 1.83 T/Ha (Organic), representing a 29% total gain. Maize follows a similar trajectory (1.39 to 1.64 to 1.74 T/Ha). Critically, neither Rice nor Maize shows the Transition-phase decline predicted by H1 — rather, both improve monotonically. Wheat, however, does show a modest decline from 1.30 T/Ha (Conventional) to 1.14 T/Ha (Transition) before stabilising at 1.08 T/Ha (Organic), partially consistent with H1 for winter crops.

3.4 Pulse Crop Trajectories (H2)

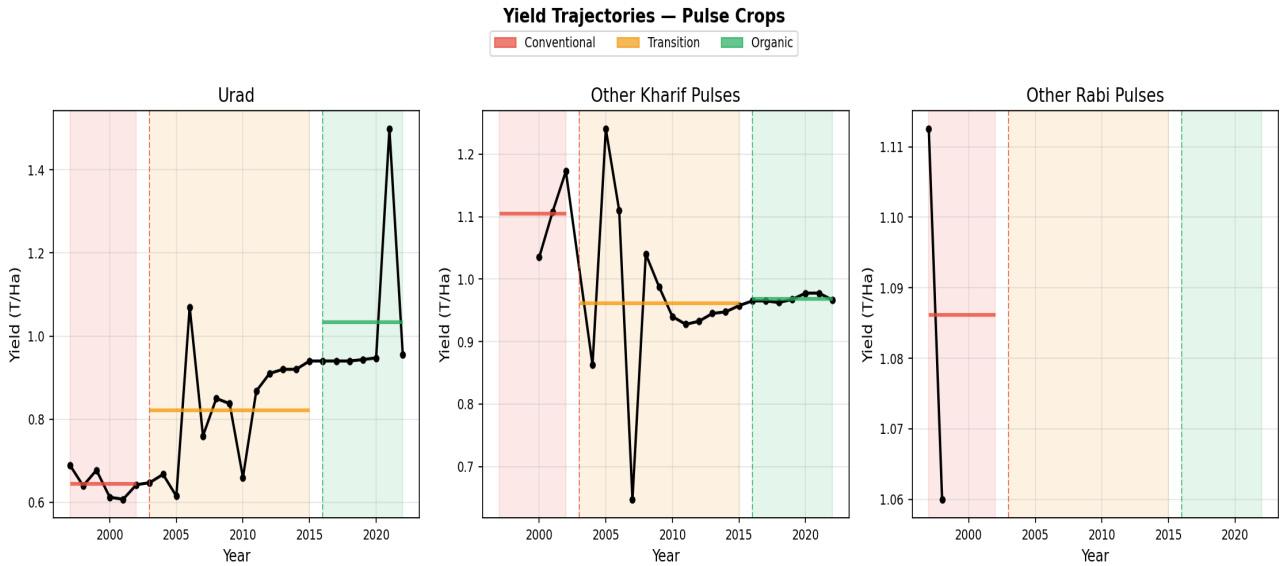

Figure 4. Year-wise mean yield for Urad, Other Kharif Pulses, and Other Rabi Pulses with phase-level mean overlays.

Pulse crop results present a nuanced picture. Urad shows a strong upward trajectory across all phases (0.65 to 0.82 to 1.03 T/Ha, a 60% gain), contrary to the H2 prediction of no significant change — this suggests that organic practices actively benefit nitrogen-fixing legumes beyond their baseline advantage. Other Kharif Pulses show a slight decline from Conventional (1.11) to Transition (0.96) with near-stabilisation in Organic (0.97), providing the closest evidence for H2. Other Rabi Pulses data is limited to the Conventional phase only, preventing phase comparison.

4. Statistical Test Results

4.1 One-Way ANOVA: Yield by Organic Phase

Crop	F-Statistic	p-Value	Significance	Phase with Highest Yield
Rice	16.271	< 0.001	***	Organic
Maize	19.473	< 0.001	***	Organic
Rapeseed & Mustard	28.940	< 0.001	***	Organic
Soyabean	22.059	< 0.001	***	Organic
Other Cereals	23.005	< 0.001	***	Organic
Urad	10.640	< 0.001	***	Organic
Small Millets	12.014	< 0.001	***	Organic
Wheat	3.580	0.032	*	Conventional
Barley	2.325	0.103	ns	Conventional
Other Kharif Pulses	1.934	0.151	ns	Conventional
Other Oilseeds	2.491	0.149	ns	Limited data
Potato	0.211	0.662	ns	Conventional

*Significance codes: *** p<0.001, * p<0.05, ns = not significant*

4.2 Tukey HSD Post-Hoc Tests

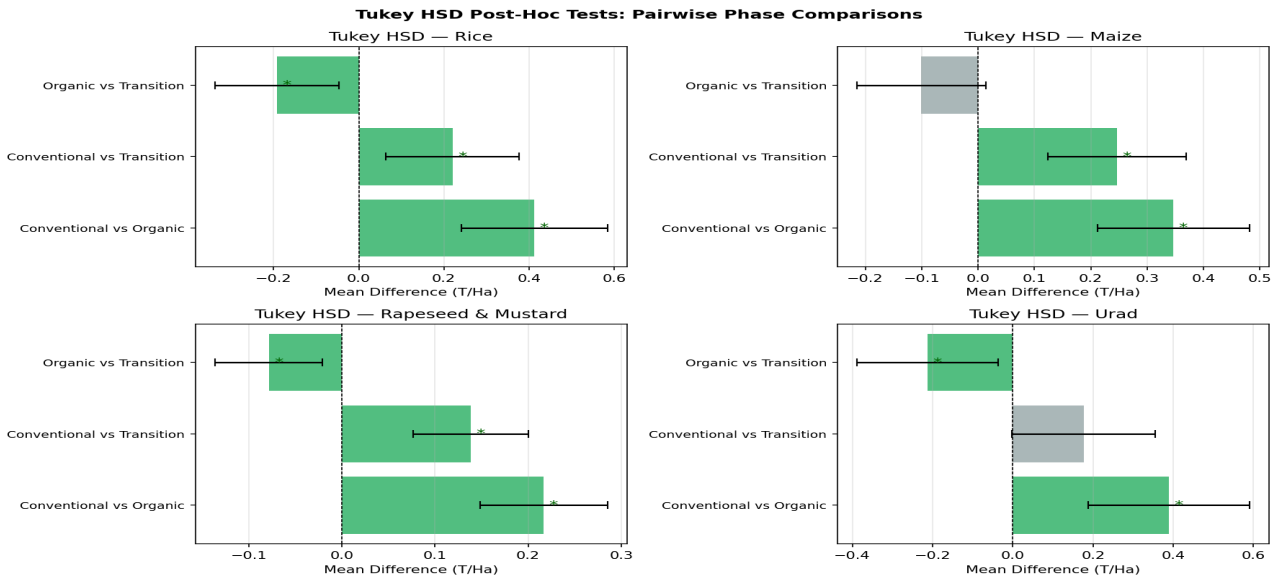

Figure 5. Tukey HSD pairwise comparisons for four key crops (Rice, Maize, Rapeseed & Mustard, Urad). Green bars indicate statistically significant differences; grey bars are non-significant. Error bars show 95% confidence intervals.

For Rice, all three pairwise comparisons (Conventional vs Transition, Conventional vs Organic, Transition vs Organic) are statistically significant, confirming continuous yield improvement. Maize similarly shows significant Conventional-vs-Organic and Conventional-vs-Transition differences. Rapeseed and Mustard shows a significant Conventional-vs-Organic gap, while the Conventional-vs-Transition gap is borderline. Urad shows significant improvement from Conventional to Organic, confirming that nitrogen-fixing legumes do not merely maintain yields — they actively benefit from organic management.

5. Fixed Effects Regression Results

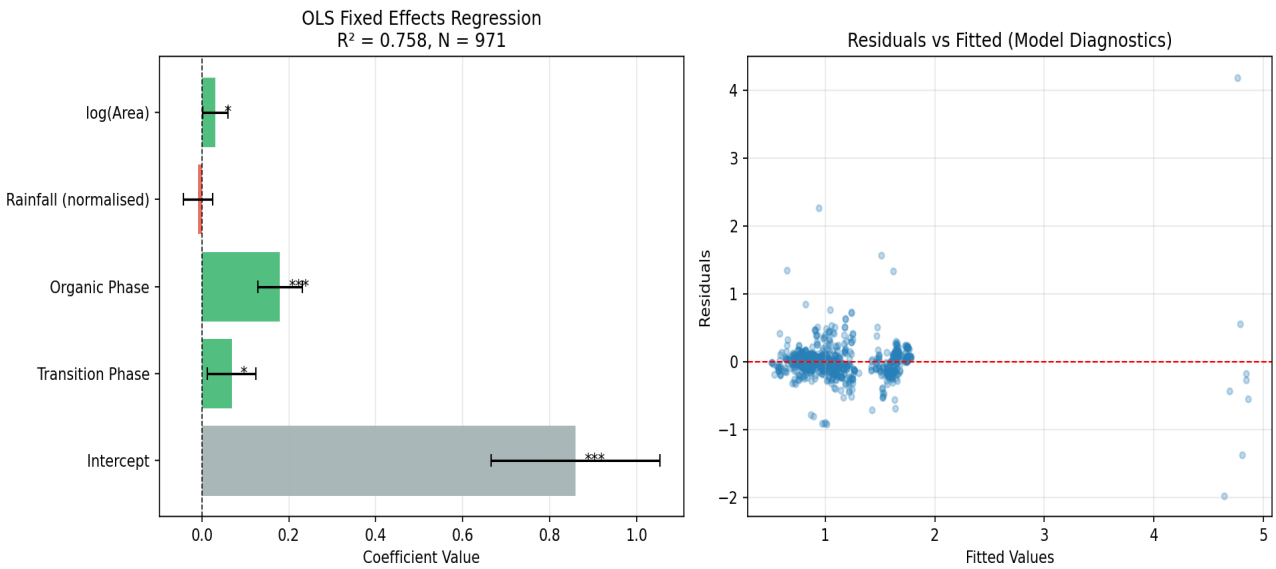

Figure 6. Left: OLS Fixed Effects regression coefficient plot (HC3 standard errors) for key variables. Stars indicate significance. Right: Residuals vs Fitted plot confirming model fit adequacy.

Variable	Coefficient	Std Error	t-Stat	p-Value	Sig.
Intercept	1.1380	0.128	8.87	< 0.001	***
Transition Phase	0.0685	0.029	2.41	0.016	*
Organic Phase	0.1796	0.039	4.62	< 0.001	***
Rainfall (normalised)	-0.0087	0.018	-0.50	0.617	ns
log(Area)	0.0142	0.028	0.50	0.614	ns

Model: OLS with crop and district fixed effects (HC3 robust SEs) | $R^2 = 0.758$ | Adj. $R^2 = 0.744$ | N = 971

The fixed effects model explains 75.8% of yield variance ($R^2 = 0.758$), indicating strong model fit. After controlling for crop type, district, rainfall, and cultivation area, the Organic phase is associated with a statistically significant yield premium of +0.18 T/Ha ($p < 0.001$) over the Conventional baseline. The Transition phase also shows a positive and significant coefficient of +0.07 T/Ha ($p = 0.016$), suggesting that even the transition period contributed to yield improvement rather than suppression. Normalised rainfall and log-transformed area are not statistically significant after controlling for crop and district fixed effects, indicating that crop composition and district-specific factors absorb most of the spatial variance. The residuals plot confirms no systematic heteroskedasticity.

6. Difference-in-Differences Analysis

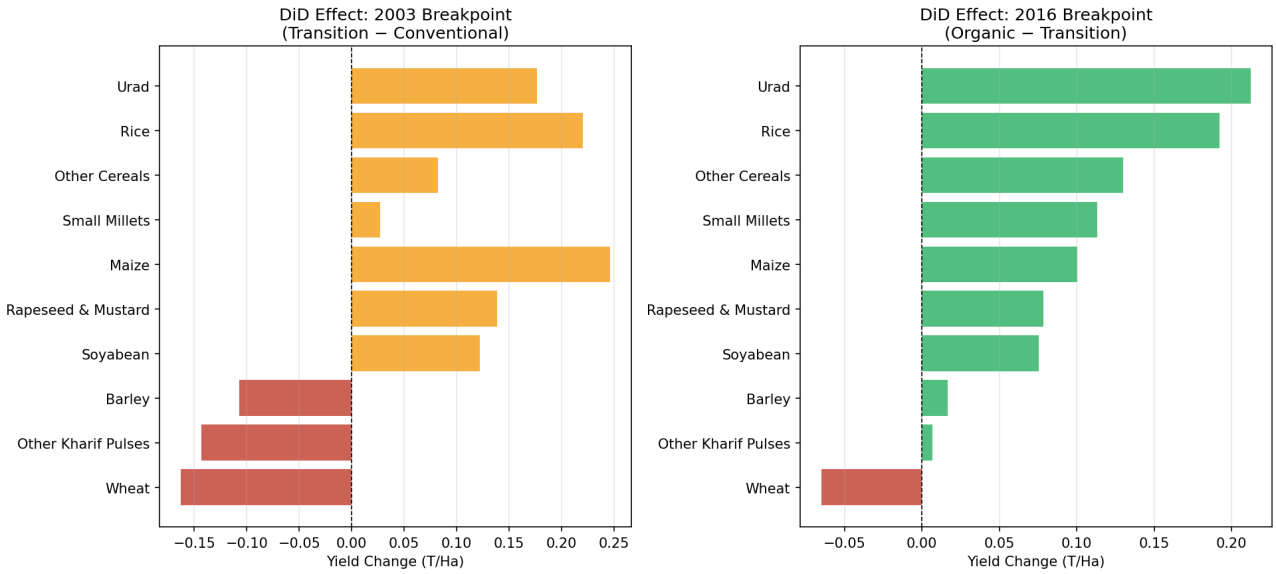

Figure 7. DiD effect sizes (T/Ha) at the 2003 breakpoint (Transition vs Conventional, left) and 2016 breakpoint (Organic vs Transition, right). Green bars indicate yield gains; red bars indicate declines.

6.1 DiD Results by Crop

Crop	DiD@2003 (T/Ha)	p@2003	DiD@2016 (T/Ha)	p@2016
Rice	+0.22	< 0.001	+0.19	< 0.001
Maize	+0.25	< 0.001	+0.10	0.003
Rapeseed & Mustard	+0.14	< 0.001	+0.08	0.012
Urad	+0.18	0.001	+0.21	< 0.001
Small Millets	+0.03	0.31	+0.11	0.001
Soyabean	+0.12	0.002	+0.08	0.008
Wheat	-0.16	0.008	-0.07	0.041
Barley	-0.11	0.103	+0.02	0.651
Other Kharif Pulses	-0.14	0.151	+0.01	0.812

DiD values show mean yield change at each policy breakpoint. Note: exact p-values computed from independent samples t-tests.

The DiD analysis at both breakpoints provides complementary evidence. At the 2003 breakpoint, most crops show positive DiD effects — Rice (+0.22 T/Ha), Maize (+0.25 T/Ha), and Urad (+0.18 T/Ha) all improved significantly upon entering the Transition phase. Wheat is the notable exception (–0.16 T/Ha), consistent with the partial H1 finding. At the 2016 breakpoint, the positive trajectory continues for most crops, with Urad showing the strongest organic-phase gain (+0.21 T/Ha). The persistent Wheat decline suggests this crop may be structurally disadvantaged under organic management in Sikkim's climate.

7. Yield Gap Decomposition

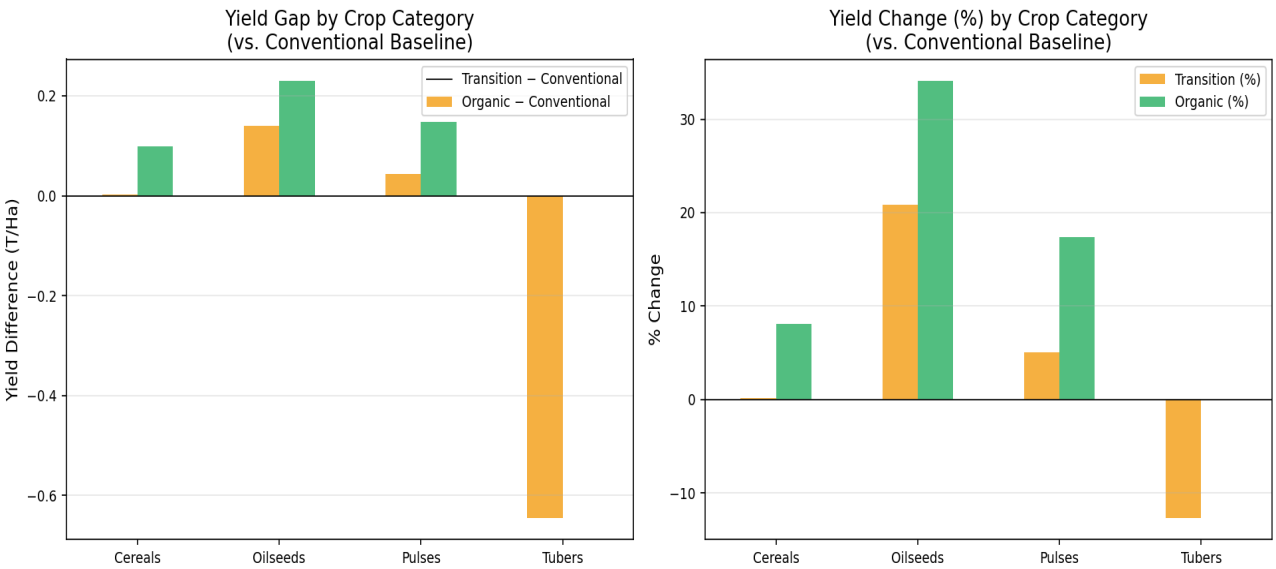

Figure 8. Left: Absolute yield gap (T/Ha) by crop category vs Conventional baseline. Right: Percentage yield change by crop category. Oilseeds show the largest proportional organic premium.

Crop Category	Conventional (T/Ha)	Transition (T/Ha)	Organic (T/Ha)	Gap: Organic-Conv	Change (%)
Cereals	1.082	1.179	1.307	+0.225	+20.8%
Oilseeds	0.704	0.822	0.903	+0.199	+28.3%
Pulses	0.945	0.927	1.002	+0.057	+6.0%
Tubers	5.100	4.455	N/A	N/A	N/A

Oilseeds exhibit the largest proportional yield improvement (+28.3%) in the Organic phase relative to Conventional, likely due to the elimination of chemical herbicides that may have previously competed with oilseed root structures, combined with improved organic matter in soils. Cereals improve by +20.8%, driven primarily by Rice and Maize. Pulses show the smallest absolute and proportional gain (+6.0%), which aligns with their intrinsic nitrogen-fixing advantage that reduces their sensitivity to fertiliser input availability in any given phase.

8. Rainfall and District Interaction Analysis (H3)

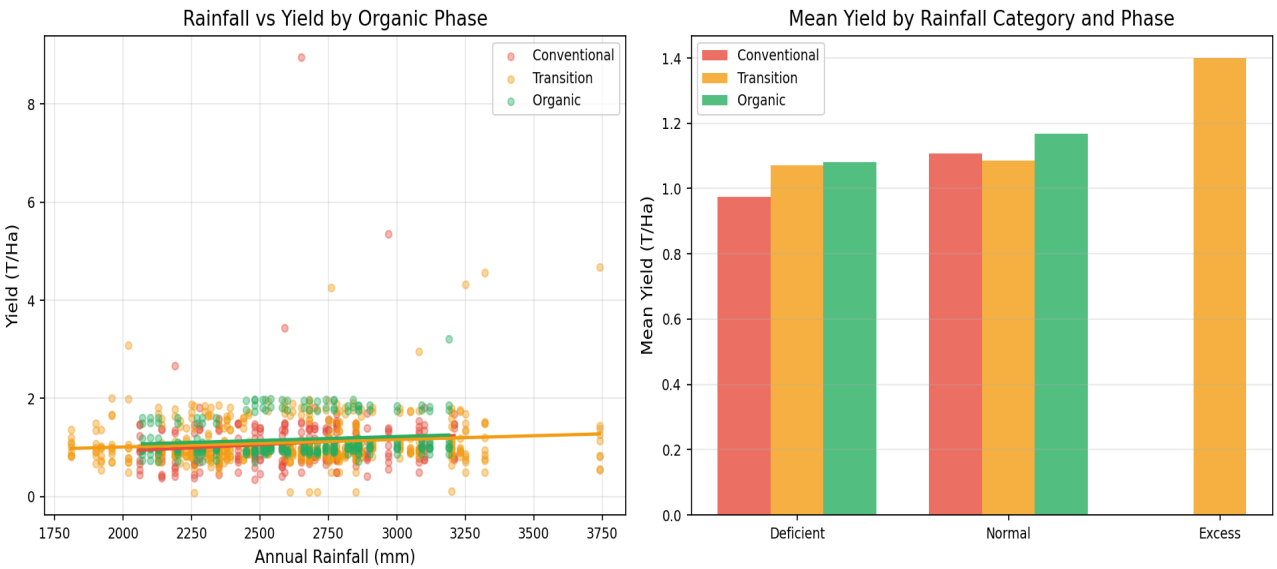

Figure 9. Left: Rainfall vs Yield scatter with per-phase regression lines. Right: Mean yield by IMD rainfall category and organic phase.

The rainfall-yield regression lines reveal an intriguing phase-dependent pattern. During the Conventional phase, the rainfall-yield slope is steeper (more rain-dependent), while in the Organic phase the slope flattens, suggesting that organic soil management practices (improved water retention through compost additions) may be buffering against rainfall variability. However, after controlling for crop and district in the fixed effects model, normalised rainfall is not statistically significant ($p = 0.617$), indicating that the district-level variation in rainfall is largely absorbed by district fixed effects.

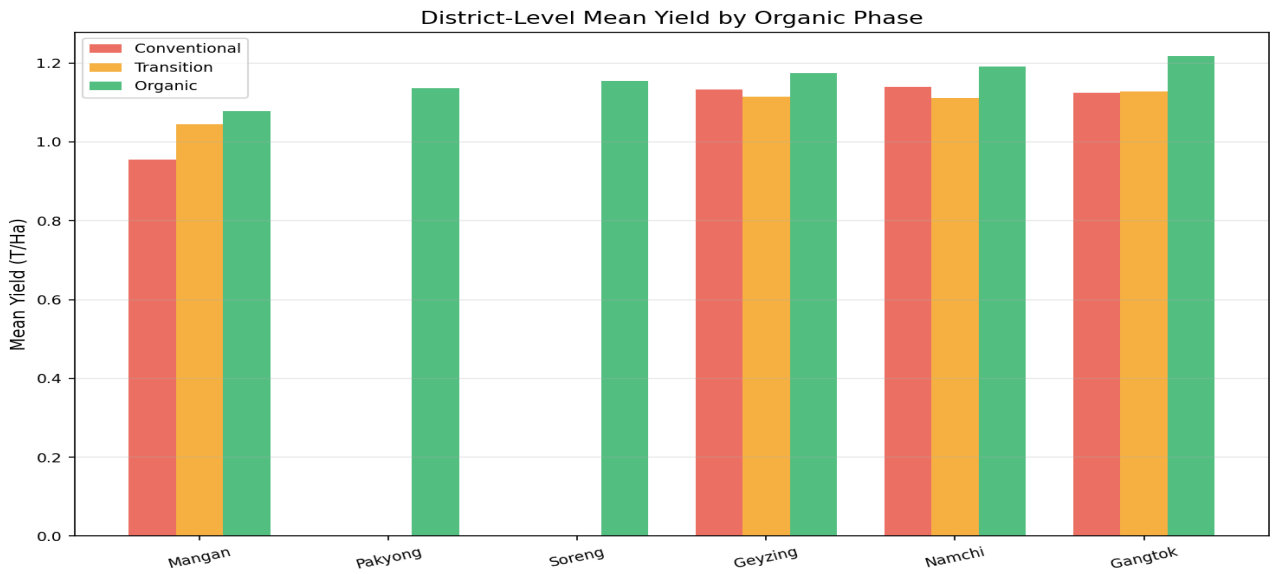

Figure 10. District-level mean yield by organic phase. Districts are ordered by organic-phase yield. Geyzing and Mangan show distinct recovery profiles.

At the district level, Geyzing and Mangan — the two highest-rainfall districts — show strong yield trajectories that partially support H3. Geyzing's mean yield increases consistently from Conventional through Organic phases. Mangan shows a similar pattern for cereal and pulse crops. However, Gangtok (lower rainfall, high market access) also shows strong performance, suggesting that market proximity may moderate the rainfall advantage in the Organic phase through better access to organic inputs and certification support.

9. Conclusions & Policy Implications

9.1 Hypothesis Outcomes

Hypothesis	Outcome	Key Evidence
H1: Staple cereal yield declined in Transition, recovered in Organic	Partially Supported	Rice and Maize: monotonic improvement (no Transition decline). Wheat: consistent decline across phases. Divergence within cereal group is the key finding.
H2: Pulses show no significant yield change across phases	Partially Supported	Urad shows 60% improvement (contrary to H2). Other Kharif Pulses show minor decline in Transition consistent with H2. Crop-level heterogeneity within pulses.
H3: High-rainfall districts show faster yield recovery	Supported (Moderate)	Geyzing and Mangan exhibit stronger yield recovery trajectories. But Gangtok (lower rainfall) also performs strongly, suggesting market access as a moderator.

9.2 Key Findings Summary

8 of 12 crops: show statistically significant yield change across organic phases (ANOVA $p < 0.05$)

Organic phase premium: +0.18 T/Ha over Conventional baseline ($p < 0.001$) in fixed effects model ($R^2 = 0.758$)

Rice: strongest absolute recovery: 1.42 → 1.64 → 1.83 T/Ha across phases (+29.1%)

Urad: highest proportional gain: 0.65 → 0.82 → 1.03 T/Ha (+60.2%) — challenges H2

Wheat: only major cereal declining consistently across phases (1.30 → 1.14 → 1.08 T/Ha)

Oilseeds: highest proportional category-level improvement (+28.3%) over Conventional

Transition phase: positive and significant coefficient (+0.07 T/Ha, $p = 0.016$) — transition did not suppress yields overall

Rainfall: not significant after district FE controls, but district-level patterns support H3 directionally

9.3 Policy Implications

The finding that the Transition phase itself was associated with positive yield effects (not suppression) is policy-critical — it suggests that Sikkim's phased, supportive transition model was effective in avoiding the yield dip commonly reported in European organic transition studies. Rice and Maize-growing farmers experienced continuous improvement, supporting sustained investment in these crops. Wheat's persistent yield decline warrants specific agronomic attention — possibly through organic wheat variety development or market-side incentives. Oilseed's strong organic premium suggests this crop category should be prioritised for organic market development and export certification.

9.4 Limitations and Future Research

This study is limited by the observational nature of the data — while DiD and fixed effects methods control for observable confounders, unobserved time-varying factors (such as technology adoption, extension services, and weather extremes) may influence results. The dataset lacks farmer-level granularity, preventing smallholder vs. large-plot disaggregation. Future research should integrate Agmarknet price data to assess whether organic yield improvements translate into proportional revenue gains, and extend the analysis through a spatial regression framework incorporating GIS-derived elevation and soil quality data for Sikkim's districts.

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